University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications Wintersemester 2025/26 Exercise 4

M.Sc. Raik Elster



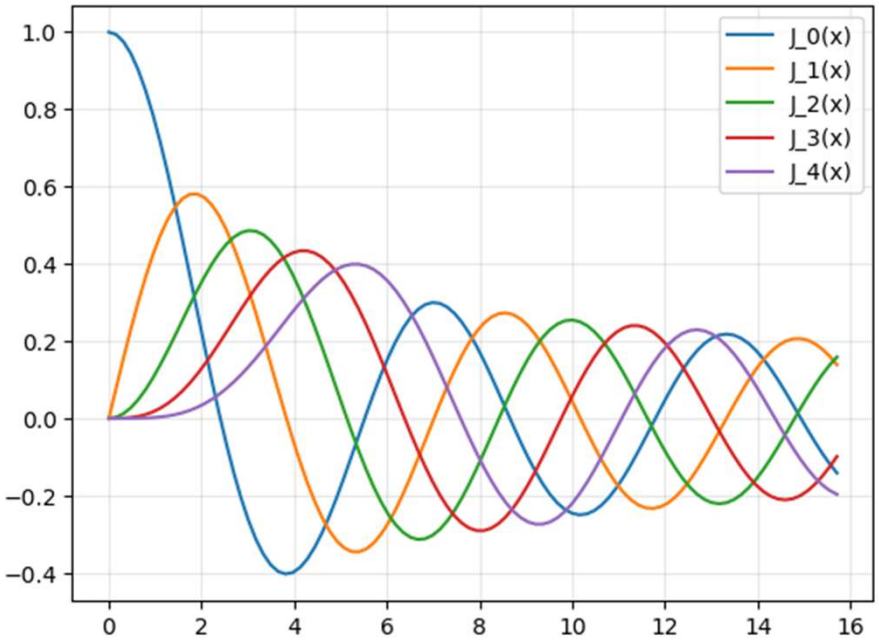
Recap:

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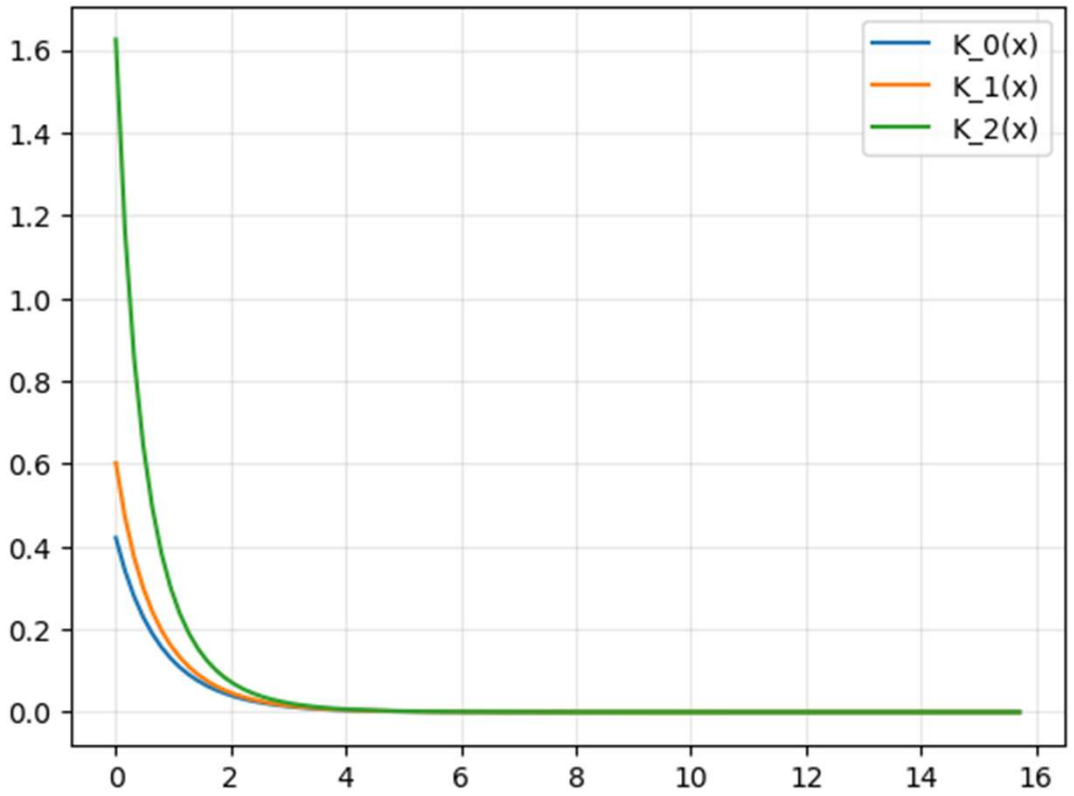
- Power Calculation for RF Systems
- Reflection and Refraction: Polarizer and Brewster angle

Optical fiber – cut off

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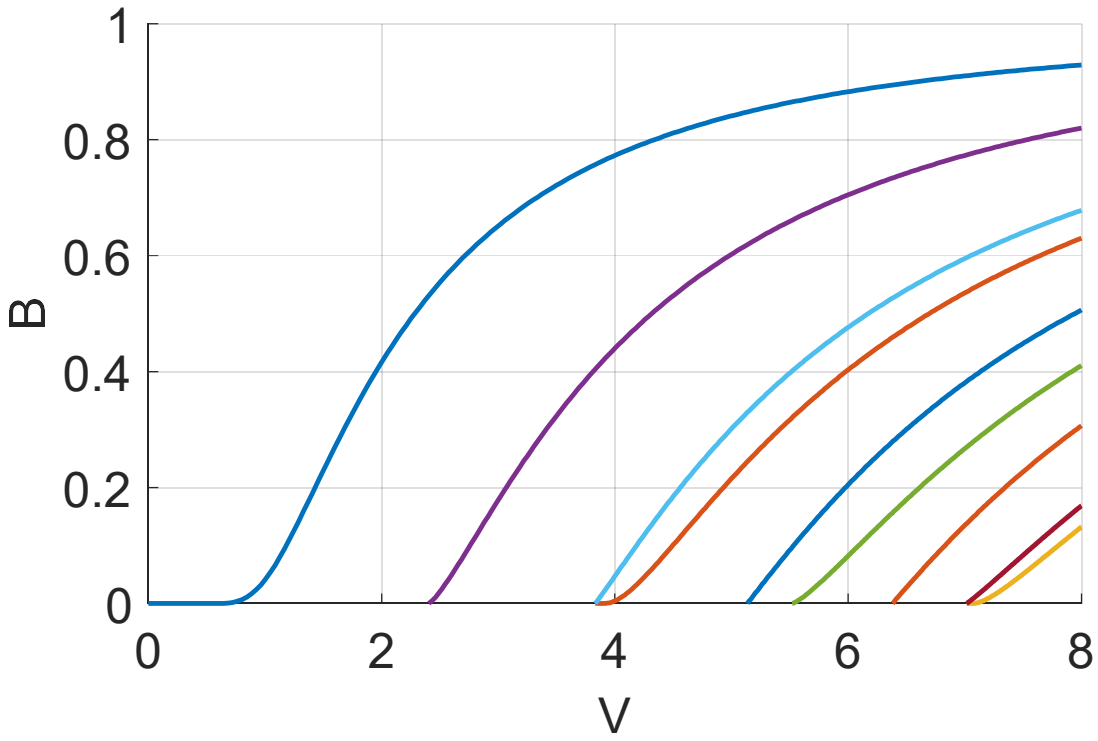


V

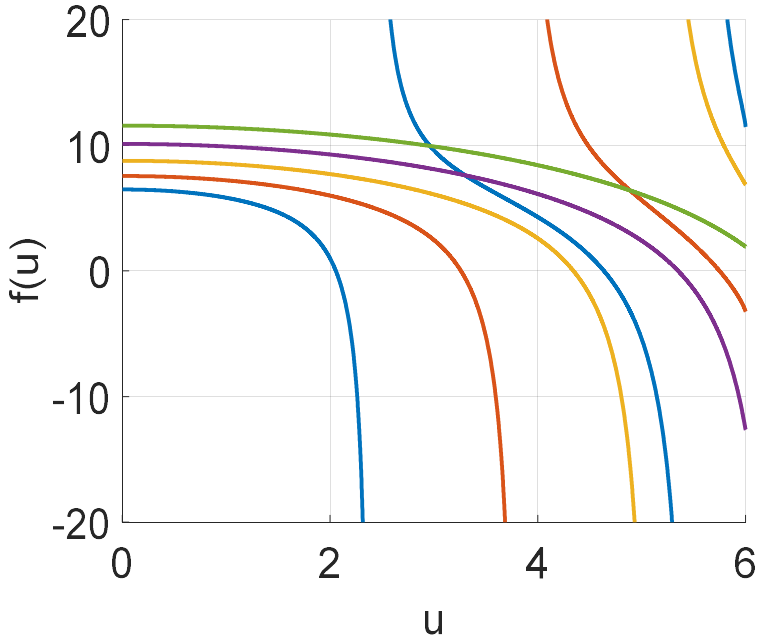

K_0(x)
K_1(x)
K_2(x)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12
14
16

Optical fiber – B-V diagram

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Optical Fiber – Characteristic equation

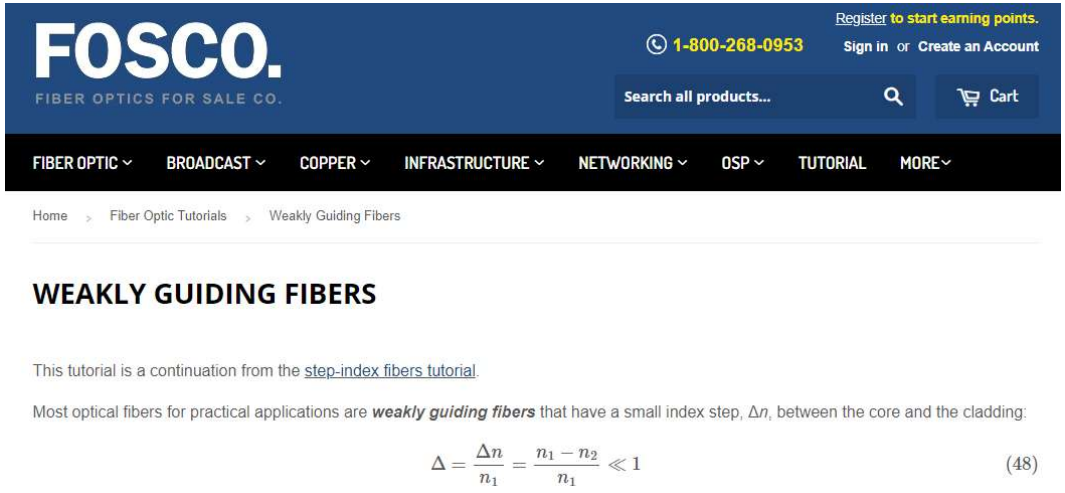


$$f(u)=\frac{uJ_{l-1}(u)}{J_l(u)}+\frac{vK_{l-1}\left(\sqrt{V^2-u^2}\right)}{K_l\left(\sqrt{V^2-u^2}\right)}$$

Optical fiber – Futher material

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https://www.fiberoptics4sale.com/blogs/wave-optics/weakly-guiding-fibers



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Home > Fiber Optic Tutorials > Weakly Guiding Fibers

WEAKLY GUIDING FIBERS

This tutorial is a continuation from the step-index fibers tutorial.

Most optical fibers for practical applications are ***weakly guiding fibers*** that have a small index step, Δn, between the core and the cladding:

$$\Delta = \frac{\Delta n}{n_1} = \frac{n_1 - n_2}{n_1} \ll 1 \qquad (48)$$

Problem 9

A step index fiber shall be used in an optical communications channel. The refractive index for the core is $n_1 = 1.457$ and for the cladding $n_2 = 1.444$. The radius of the core $a = 6.1$ µm and the used wavelength is 1.55 µm.

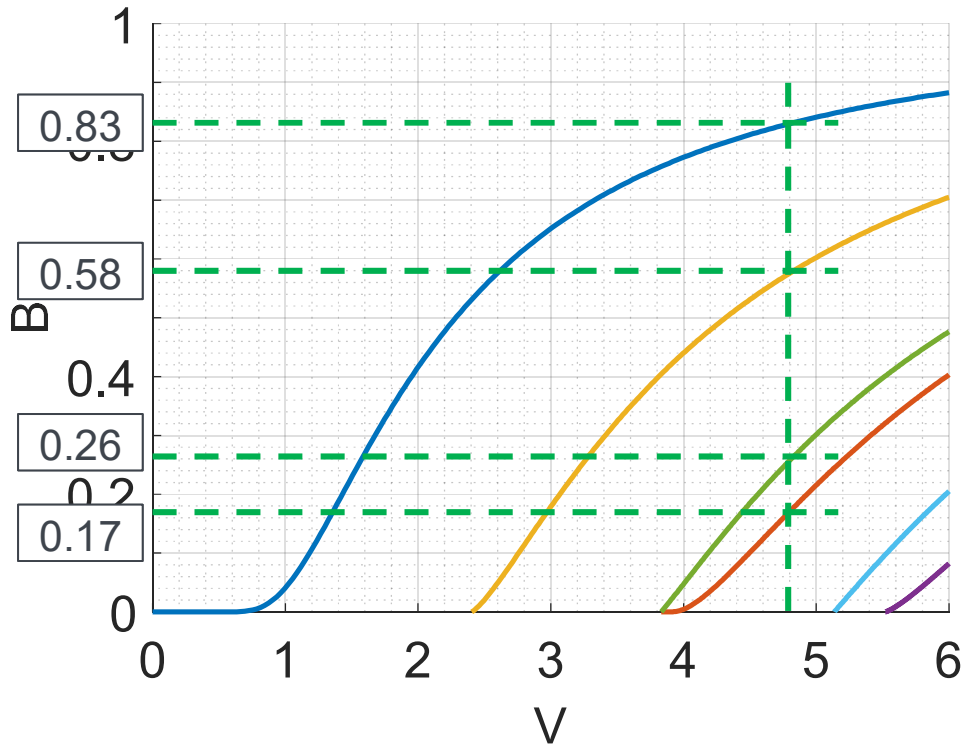
a) Find the effective refractive indices for all modes that the step index fiber supports with the help of the B-V diagram. What are the corresponding incident angels of those modes at the core-cladding interface?

b) Calculate the waveguide dispersion group delay for all supported modes by using the d(VB)/dV diagram. Which mode is the slowest and which one the fastest?

Problem 9

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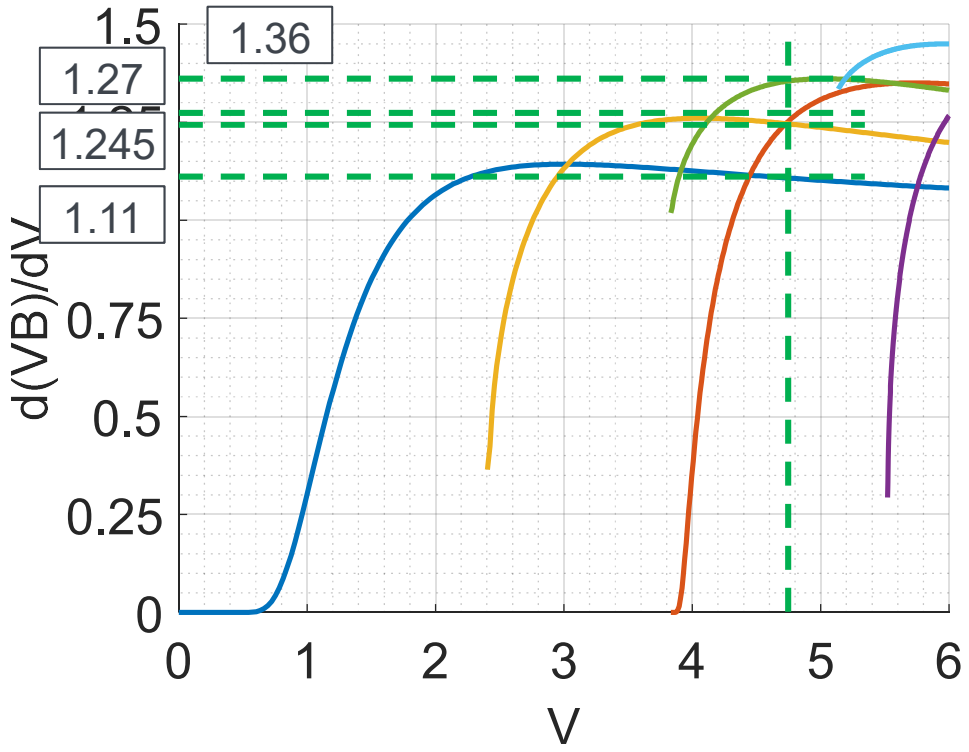
a)



Problem 9

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b)



Problem 9

The length of the used fiber is 2.3 km. Only the fastest and the slowest mode have to be considered from now on.

c) What is the time delay between the modes?

d) Sketch the signal at the output of the fiber for the symbol rates of 10 MBd and 30 MBd with the following NRZ coded symbols. The power is equally distributed on both modes.

n	1	2	3	4	5	6	7	8	9	10
s_n	0	1	0	1	0	0	1	1	0	0

e) What is the bit error rate (BER) for both symbol rates, if the threshold is a hysteresis with the value 0.6 for 0-to-1 transition and 0.4 for 1-to-0 transition? The sampling takes place in the center of a symbol period.